REMY PROSPECT AI

Product Requirements Document (PRD)

Document version: 1.0
Status: Product specification for implementation
Product type: Standalone, responsive web application
Primary owner: Platform creator
Initial users: One personal account, with controlled support for additional users
Target markets: Nigeria and international markets
Initial budget: $0
Primary objective: Automatically discover, research, rank, and notify the owner about relevant development opportunities and prospective clients.

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1. EXECUTIVE SUMMARY

Remy Prospect AI is a personal AI-powered client acquisition and opportunity research platform designed to help an AI-assisted developer find paid work without depending entirely on freelance marketplaces that require paid credits or subscriptions.

The platform will continuously search permitted public sources for freelance projects, job listings, company announcements, and businesses that may need software development services. It will collect relevant information, investigate each opportunity, assess its relevance to the owner's capabilities, assign a transparent lead score, and deliver notifications through email and the application dashboard.

Unlike a conventional job board, Remy Prospect AI will combine two complementary acquisition strategies:

1. Active opportunity discovery: Identify clients who have explicitly advertised a project, requested a developer, or published a relevant hiring announcement.
2. Proactive business prospecting: Discover companies that may benefit from the owner's services, research their publicly available business information, and identify evidence of potential technical needs.

Both strategies will receive equal priority.

The platform will initially support opportunities involving:

- AI-assisted website development.
- AI-assisted mobile application development.
- API integrations and third-party service integrations.
- AI solutions, chatbots and AI-powered application features.
- Web applications, dashboards and SaaS products.
- Website maintenance, redesigns, bug fixes and performance improvements.
- Related software development, automation and technical services.

The application will be a separate product, not an implementation inside the existing Remy AI application. It will work through a responsive web interface on mobile phones, tablets and laptops.

The platform will use a free-first architecture, with optional paid integrations introduced only when justified.

1.1 Product vision

Build a reliable personal assistant that helps its owner identify and pursue relevant paid development opportunities, reducing the time spent manually searching job websites, researching companies and deciding which prospects deserve attention.

1.2 Product mission

Turn scattered public opportunity information into a structured, prioritized and actionable pipeline of potential clients.

1.3 Core product promise

The system should be able to answer the following questions:

- Who may need my development services?
- What project or technical problem do they have?
- Where was the opportunity published?
- What evidence supports this opportunity?
- How well does it match my skills?
- Is there evidence of budget or commercial intent?

- How can I legitimately contact the prospect?
- Why should I prioritize this opportunity?
- What should I do next?
- Have I already reviewed or contacted this prospect?

The product must provide evidence and original source links rather than unsupported AI-generated claims.

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2. PROBLEM STATEMENT

2.1 Current challenges

The owner has completed training in AI-assisted web development and application development and is seeking paid development work.

Traditional freelance platforms can impose financial barriers, including paid proposal credits, subscriptions or other access requirements. This makes it difficult to compete when starting without capital.

Searching manually across different platforms also introduces several problems:

- Opportunities are distributed across many websites.
- Relevant job posts may be difficult to discover.
- New opportunities appear at different times.
- Some job listings are poorly matched to the developer's current skills.
- Company research takes time.
- Contact details may be unavailable or difficult to verify.
- It is easy to overlook promising prospects.
- Repeated searches can produce duplicate results.
- Follow-ups and applications can become disorganized.
- A developer may spend time on low-quality leads rather than promising opportunities.

2.2 Proposed solution

Remy Prospect AI will centralize opportunity discovery, research, scoring, notifications and follow-up tracking in one application.

The system will automatically perform the research and prioritization work wherever technically and legally permitted.

The owner will be able to concentrate on reviewing opportunities, contacting suitable prospects, delivering projects and improving the product based on actual results.

2.3 Intended outcome

The platform should reduce manual research effort and help the owner maintain a consistent pipeline of potential clients.

The system cannot guarantee employment, contracts, responses, or revenue. Success will be measured by the quality of discovered opportunities and their progression into real business conversations.

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3. PRODUCT OBJECTIVES

3.1 Primary objectives

1. Discover relevant development opportunities automatically.
2. Find prospective businesses in Nigeria and international markets.
3. Research opportunities and companies using publicly available evidence.
4. Match opportunities against the owner's services, skills and portfolio.
5. Rank opportunities using a transparent 0–100 scoring system.
6. Notify the owner through both email and in-app dashboard notifications.
7. Maintain a searchable database of leads and research findings.
8. Track applications, outreach, replies, follow-ups and outcomes.
9. Minimize initial operating costs.
10. Respect the access restrictions, terms and privacy requirements of external sources.

3.2 Secondary objectives

- Identify recurring demand for particular development services.
- Track which discovery sources generate useful opportunities.
- Learn which lead characteristics correlate with replies and paid projects.
- Allow additional permitted data sources to be added without rebuilding the

application.
- Support additional users and more sophisticated administration if needed later.

3.3 Non-objectives for the MVP

The first release will not:

- Guarantee paid work or a minimum number of leads.
- Bypass marketplace paywalls or paid access restrictions.
- Extract private information from job posters.
- Circumvent authentication, CAPTCHAs or anti-bot mechanisms.
- Automatically apply for every job.
- Automatically send unsolicited messages in bulk.
- Purchase subscriptions or use paid APIs without authorization.
- Build a separate Android APK.
- Replace the owner's professional judgment.
- Require a complex multi-tenant enterprise architecture.
- Depend on paid AI inference to perform basic lead management.

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4. TARGET USERS AND PERSONAS

4.1 Primary persona: Platform owner

The initial owner is an AI-assisted developer who wants to find paid work while starting with no available budget for expensive job-search subscriptions.

The owner needs to:

- Configure services and skills.
- Search Nigerian and international markets.
- Receive new opportunities without constantly monitoring websites.
- Review leads on a phone or laptop.
- Understand why a lead is recommended.
- Identify legitimate contact channels.
- Track applications and business conversations.
- Manage the application's configuration and security.

4.2 Secondary persona: Administrator

An administrator may help manage the discovery system and review leads.

The administrator should have operational privileges without automatically gaining access to all security-critical controls.

4.3 Future persona: Additional standard user

A future user may maintain their own profile, service preferences, lead pipeline and notifications.

This capability is secondary to the initial single-owner experience.

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5. SERVICE CATEGORIES AND SEARCH CONFIGURATION

The owner must be able to configure the services the platform searches for.

5.1 Initial service categories

A. AI-assisted web development

Search for projects involving:

- Business websites.
- Landing pages.
- Portfolio websites.
- E-commerce websites.
- SaaS applications.
- Admin dashboards.
- Customer portals.
- Responsive web applications.
- Website modernization.
- Website redesigns.

B. Mobile application development

Search for:

- Android applications.
- Cross-platform mobile applications.
- Mobile application MVPs.
- AI-powered mobile applications.
- Mobile API integrations.
- Application maintenance and debugging.

C. API integrations

Search for:

- REST APIs.
- Third-party service integrations.
- Payment gateway integrations.
- Authentication integrations.
- Webhooks.
- Data synchronization.
- Workflow automation.
- Backend service integrations.

D. AI solutions

Search for:

- AI chatbots.
- LLM integrations.
- AI-powered search.
- Retrieval-augmented generation.
- AI agents.
- Document assistants.
- Business workflow automation.
- AI-enhanced existing applications.

E. Related development services

Search for:

- Bug fixing.
- Performance optimization.
- Frontend development.

- Backend development.
- Database design.
- Application maintenance.
- Feature development.
- Technical consulting within the owner's demonstrated capabilities.

5.2 Search configuration

The owner must be able to configure:

- Service categories.
- Keywords and alternative phrases.
- Excluded keywords.
- Target countries and regions.
- Remote, hybrid or location-specific work.
- Job type, such as freelance, contract, project-based or other relevant work.
- Minimum lead score.
- Search frequency.
- Source enablement.
- Notification preferences.

The default market preference is Nigeria and international opportunities equally.

The system should not exclude an opportunity merely because a country is unfamiliar. It should evaluate service fit, eligibility, language, location requirements and other relevant constraints.

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6. PRODUCT ARCHITECTURE

6.1 Architecture overview

The application will use a modular, provider-independent architecture.

The principal components are:

1. Responsive web frontend.
2. Authentication and authorization.
3. Application backend.

4. PostgreSQL database.
5. Discovery and source integration layer.
6. Research and AI analysis layer.
7. Lead scoring engine.
8. Scheduled job processor.
9. Notification delivery system.
10. Administrative control panel.
11. Logging, monitoring and audit facilities.

6.2 Recommended technology stack

Frontend

- React.
- TypeScript.
- Vite.
- A consistent, accessible component library.
- Responsive layouts for mobile and desktop.

Backend

- Supabase Auth.
- Supabase PostgreSQL.
- Supabase Edge Functions or another appropriate server-side execution environment.
- Scheduled triggers or a supported external scheduler.

AI layer

- A provider abstraction for AI model integration.
- Structured JSON output with validation.
- Rule-based fallbacks when an AI provider is not configured.
- Optional model APIs when usage credits or a budget are available.

Discovery layer

- Public web search APIs.
- Public feeds where available.
- Official company websites.
- Permitted public job feeds and documented APIs.

- Manual URL import as a fallback.

Notifications

- Persistent dashboard notifications.
- Resend or another suitable transactional email provider.

Source control and deployment

- GitHub.
- A suitable free-tier hosting provider.
- Supabase free tier where its limits and terms are sufficient.

6.3 Architectural principles

- Keep provider-specific integrations separate from core application logic.
- Avoid hard dependencies on any single search or AI provider.
- Keep credentials on the server.
- Use a common normalized lead format across all discovery sources.
- Make scheduled jobs idempotent.
- Store source evidence separately from AI-generated conclusions.
- Preserve functionality when optional integrations are unavailable.
- Avoid unnecessary complexity in the first release.

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7. FUNCTIONAL REQUIREMENTS

7.1 Authentication and account management

AUTH-001 — Sign-up and provisioning

The application must support secure account creation through a controlled process.

Public registration should be disabled by default or restricted to explicitly authorized registration.

AUTH-002 — Sign-in

Users must be able to authenticate using email and password.

AUTH-003 — Sign-out

Users must be able to sign out from the interface. The application must invalidate or revoke the applicable session using the supported authentication mechanism and clear client-side authenticated state.

AUTH-004 — Password recovery

Users must be able to request a password reset through a secure email workflow.

AUTH-005 — Email verification

Support email verification where enabled in the authentication configuration.

AUTH-006 — Protected routes

Unauthenticated users must not access private application routes or protected data.

AUTH-007 — Session management

Support session persistence, expiry and revocation using the chosen authentication provider's supported capabilities.

AUTH-008 — Profile management

Users must be able to update their permitted profile fields, service preferences and notification settings.

AUTH-009 — Account recovery

Document a secure recovery process for the platform owner, including recovery of Super Admin access without exposing a public privilege-escalation endpoint.

AUTH-010 — Privileged authentication

Require stronger protection for privileged accounts, including MFA where supported.

7.2 Opportunity discovery

DISC-001 — Scheduled search

The system must run searches automatically without requiring the owner to keep the browser open.

DISC-002 — Multiple sources

Support multiple discovery sources through a common provider interface.

DISC-003 — Active opportunity discovery

Search for publicly available job posts, project requests, hiring announcements and requests for technical services.

DISC-004 — Business prospecting

Find companies that may need the owner's services and record the evidence supporting the prospecting hypothesis.

DISC-005 — Equal discovery priority

Neither active opportunities nor proactive prospecting should be permanently deprioritized. Search allocation must be configurable and balanced by default.

DISC-006 — Source URLs

Every externally discovered lead must include its original source URL when available.

DISC-007 — Manual import

Allow the owner to paste a URL or enter opportunity details manually.

DISC-008 — Source status

Administrators must be able to enable or disable sources and inspect their last successful execution.

DISC-009 — Deduplication

Repeated discovery of the same opportunity must not create unlimited duplicate records.

DISC-010 — Incremental updates

When a previously discovered listing changes meaningfully, update the record and retain appropriate history.

DISC-011 — Geographic targeting

Support Nigerian and international opportunities.

DISC-012 — Search terms

Support editable keywords, alternative phrases and exclusions.

DISC-013 — Source restrictions

Do not perform unauthorized scraping, bypass source restrictions or extract information prohibited by the applicable source's terms.

DISC-014 — Run history

Record discovery runs, execution times, results, failures and retry attempts.

7.3 Company research

RES-001 — Company identification

Extract the company or organization name where available.

RES-002 — Company profile

Summarize publicly available information relevant to the opportunity.

RES-003 — Website review

Where access is permitted, inspect public website pages for relevant business context and technical observations.

RES-004 — Evidence storage

Record observations with source URLs and timestamps.

RES-005 — Contact discovery

Identify publicly published business contact channels where permitted.

RES-006 — Contact verification

Distinguish verified details, unverified details and missing information.

RES-007 — Research confidence

Assign a confidence assessment based on source quality, corroboration and freshness.

RES-008 — Hypothesis labeling

Clearly label potential needs as hypotheses unless supported by explicit evidence.

RES-009 — Research summary

Provide a concise summary of the prospect, potential need, available evidence and recommended next step.

RES-010 — No fabricated facts

The AI must never invent contact information, budgets, hiring intentions, company facts or job requirements.

7.4 AI matching and scoring

SCORE-001 — Service matching

Compare the opportunity's requirements against the owner's selected services.

SCORE-002 — Skills matching

Compare required technologies and skills against the owner's profile.

SCORE-003 — Budget assessment

Evaluate available budget information and commercial-intent signals without treating unknown information as proof of low value.

SCORE-004 — Freshness assessment

Evaluate how recently the opportunity was published or updated.

SCORE-005 — Source confidence

Evaluate source credibility and whether important details are supported by evidence.

SCORE-006 — Total score

Generate a score from 0 to 100.

SCORE-007 — Explanation

Show the score components and explain why the lead received its score.

SCORE-008 — Threshold configuration

Allow the owner to configure the minimum score required for automatic notifications.

SCORE-009 — Re-scoring

Recalculate scores when relevant information or scoring rules change.

SCORE-010 — Model independence

Core lead storage and rule-based scoring must work without a paid AI model API.

7.5 Lead management

LEAD-001 — Lead list

Display discovered opportunities in a searchable, filterable list.

LEAD-002 — Lead detail

Display the complete available description, source, research summary, score, evidence and contact channels.

LEAD-003 — Status management

Support at least the following statuses:

- New.
- Reviewed.
- Shortlisted.
- Contacted.
- Applied.
- Replied.
- Interviewing.
- Won.
- Lost.
- Dismissed.

LEAD-004 — Notes

Allow the owner to record private notes.

LEAD-005 — Follow-up dates

Allow the owner to schedule and review follow-up reminders.

LEAD-006 — Duplicate visibility

Identify duplicates and link related records where useful.

LEAD-007 — Source navigation

Allow the owner to open the original listing or company website.

LEAD-008 — Outcome tracking

Record whether the lead resulted in a reply, interview, project or contract.

LEAD-009 — Export

Provide an export option for the owner's lead records, subject to applicable data restrictions.

7.6 Notifications

NOTIF-001 — Dashboard notifications

Every qualifying new opportunity must generate a persistent in-app notification.

NOTIF-002 — Email notifications

Every qualifying new opportunity must be eligible for an email alert when email notifications are enabled.

NOTIF-003 — Dual delivery

The system must support both email and dashboard notifications simultaneously.

NOTIF-004 — Notification content

Include the opportunity title, company where known, match score, short explanation and original source link.

NOTIF-005 — Unread count

Display the number of unread dashboard notifications.

NOTIF-006 — Read state

Allow notifications to be marked as read or dismissed.

NOTIF-007 — Delivery status

Record whether email delivery was queued, accepted, failed or requires retry.

NOTIF-008 — Duplicate prevention

Avoid sending repeated notifications for the same unchanged opportunity.

NOTIF-009 — Threshold

Default to a 65/100 score threshold, configurable by the owner.

NOTIF-010 — Failure independence

Email failure must not delete or suppress the dashboard notification.

NOTIF-011 — Notification preferences

Allow email and dashboard preferences to be configured separately.

NOTIF-012 — Delivery retries

Retry transient failures using bounded exponential backoff.

NOTIF-013 — Email verification

Do not attempt production notifications to an unverified or unconfigured destination where verification is required.

7.7 Administration

ADMIN-001 — Role-based access control

Support "super_admin", "admin" and "user" roles.

ADMIN-002 — Super Admin dashboard

Provide privileged configuration and monitoring pages.

ADMIN-003 — User management

Allow authorized administrators to view accounts, manage permitted account

states and revoke access.

ADMIN-004 — Role management

Only Super Admin-authorized server-side operations may grant or revoke privileged roles.

ADMIN-005 — Source management

Allow authorized administrators to configure approved sources.

ADMIN-006 — Job monitoring

Display discovery-run history, errors and retry status.

ADMIN-007 — Notification monitoring

Display email delivery and dashboard notification status.

ADMIN-008 — Usage monitoring

Track provider usage and configured limits.

ADMIN-009 — Audit logs

Record security-relevant administrative operations.

ADMIN-010 — Configuration management

Allow authorized updates to search schedules, scoring thresholds and other permitted system settings.

ADMIN-011 — Secret protection

Do not expose API keys or other secrets through the administrative interface.

ADMIN-012 — Role separation

An Admin must not promote themselves or another user to Super Admin.

ADMIN-013 — Owner recovery

Maintain a documented, secure procedure for recovering the initial owner's privileged access.

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8. ADMINISTRATIVE PERMISSION MATRIX

Capability| Super Admin| Admin| User
View personal dashboard| Yes| Yes| Yes
View and manage permitted leads| Yes| Yes| Own leads
Manage discovery sources| Full control| Approved sources| No
View system run history| Yes| Yes| No
Retry failed jobs| Yes| Permitted jobs| No
Configure global scoring| Yes| Limited| Personal threshold only
Manage notification delivery| Yes| Operational access| Own preferences
Manage user accounts| Yes| Limited, if authorized| No
Grant Admin role| Yes| No| No
Grant Super Admin role| Restricted trusted process| No| No
Change protected security settings| Yes| No| No
View audit logs| Yes| Limited, if authorized| No
Access secrets| Through approved server-side secret management only| No direct access| No

All permissions must be enforced on the server and in database policies. Frontend navigation visibility is not a security boundary.

The initial account will be provisioned as Super Admin through a trusted, documented bootstrap procedure.

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9. USER INTERFACE AND INFORMATION ARCHITECTURE

9.1 Public routes

- "/login"
- "/signup" — only when registration is enabled.
- "/forgot-password"

- "/reset-password"

9.2 Authenticated application routes

- "/dashboard"
- "/opportunities"
- "/opportunities/:id"
- "/companies"
- "/research"
- "/notifications"
- "/settings/profile"
- "/settings/services"
- "/settings/search"
- "/settings/notifications"
- "/settings/security"

9.3 Administrative routes

- "/admin"
- "/admin/users"
- "/admin/roles"
- "/admin/sources"
- "/admin/jobs"
- "/admin/notifications"
- "/admin/usage"
- "/admin/audit-logs"
- "/admin/settings"

Every administrative route must validate the current user's role on the server before returning protected information or performing a privileged action.

9.4 Main dashboard

The dashboard should display:

- Total discovered opportunities.
- New opportunities.
- High-scoring leads.
- Leads awaiting review.
- Applications or outreach in progress.

- Replies and interviews.
- Won and lost opportunities.
- Recent discovery activity.
- Recent notifications.
- Last successful discovery run.
- Source health and failure indicators.

All metrics must be based on actual stored data rather than hardcoded demonstration figures.

9.5 Opportunity list

The list should support filters for:

- Service category.
- Country or region.
- Active job versus proactive prospect.
- Lead score.
- Publication date.
- Source.
- Budget information.
- Lead status.
- Contact availability.
- Reviewed or unread state.

9.6 Mobile usability

The mobile interface must:

- Adapt to narrow screens.
- Use touch-friendly controls.
- Keep navigation accessible.
- Avoid horizontally overflowing tables.
- Display lead scores and source links clearly.
- Support notification review without requiring desktop access.

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10. DATABASE REQUIREMENTS

The database will use PostgreSQL through Supabase.

All user-owned data must be protected by Row Level Security. Server-only operational tables must not be exposed through unrestricted client access.

10.1 "profiles"

Purpose: Store account and professional preferences.

Fields:

- "id" — UUID, matching the authenticated user.
- "display_name" — nullable text.
- "email" — account email as appropriate.
- "skills" — structured list of skills.
- "services" — structured list of offered services.
- "portfolio_urls" — list of URLs.
- "target_markets" — list of regions or countries.
- "experience_summary" — optional text.
- "preferences" — validated JSON settings.
- "created_at".
- "updated_at".

10.2 "user_roles"

Purpose: Maintain trusted application roles.

Fields:

- "id".
- "user_id".
- "role".
- "granted_by".
- "created_at".
- "updated_at".

Constraints:

- Role must be one of the approved values.
- Prevent duplicate active role assignments where inappropriate.

- Ordinary users cannot update their own role.
- Privileged changes must use a trusted server-side function.
- Role changes must be audited.

10.3 "discovery_sources"

Purpose: Configure and monitor sources.

Fields:

- "id".
- "name".
- "source_type".
- "provider_key".
- "enabled".
- "configuration".
- "last_success_at".
- "last_failure_at".
- "created_at".
- "updated_at".

Credentials must not be stored in client-readable configuration fields.

10.4 "search_runs"

Purpose: Track scheduled and manual discovery.

Fields:

- "id".
- "source_id".
- "started_at".
- "finished_at".
- "status".
- "items_found".
- "items_created".
- "items_updated".
- "error_code".
- "error_summary".
- "retry_count".

- "created_at".

10.5 "leads"

Purpose: Store normalized opportunities.

Fields:

- "id".
- "profile_id".
- "source_id".
- "search_run_id".
- "external_id".
- "source_url".
- "title".
- "company_name".
- "description".
- "opportunity_type".
- "service_categories".
- "required_skills".
- "country".
- "region".
- "work_location_type".
- "published_at".
- "discovered_at".
- "last_seen_at".
- "budget_min".
- "budget_max".
- "currency".
- "contact_url".
- "public_email".
- "contact_verification_status".
- "status".
- "content_hash".
- "created_at".
- "updated_at".

Budget and contact fields must remain nullable. Unknown values must not be fabricated.

10.6 "lead_evidence"

Purpose: Store the research supporting a lead.

Fields:

- "id".
- "lead_id".
- "evidence_type".
- "source_url".
- "observation".
- "confidence".
- "observed_at".
- "created_at".

10.7 "lead_scores"

Purpose: Store explainable scoring results.

Fields:

- "id".
- "lead_id".
- "relevance_score".
- "need_score".
- "budget_score".
- "recency_score".
- "source_confidence".
- "total_score".
- "explanation".
- "model_version".
- "scoring_version".
- "created_at".

10.8 "notifications"

Purpose: Store persistent notification records.

Fields:

- "id".
- "profile_id".
- "lead_id".
- "channel".
- "event_type".
- "title".
- "message".
- "read_at".
- "dismissed_at".
- "delivery_status".
- "attempt_count".
- "last_attempt_at".
- "sent_at".
- "idempotency_key".
- "created_at".

10.9 "outreach_logs"

Purpose: Track professional outreach and results.

Fields:

- "id".
- "lead_id".
- "channel".
- "action".
- "sent_at".
- "response_status".
- "follow_up_at".
- "notes".
- "created_at".

The application will not automatically send messages in the MVP. This table records actions performed or explicitly recorded by the owner.

10.10 "audit_logs"

Purpose: Record administrative and security-sensitive actions.

Fields:

- "id".
- "actor_user_id".
- "action".
- "target_type".
- "target_id".
- "metadata".
- "created_at".

Avoid storing credentials, authentication tokens or unnecessary personal information in audit logs.

10.11 Supporting infrastructure

The implementation may also include:

- A notification outbox table for reliable delivery.
- A scheduled-task lock or execution table.
- A source permission and terms registry.
- A provider-usage ledger.
- A company table if normalized company profiles become necessary.

These additions should be introduced only where they improve reliability, security or data consistency.

10.12 Data integrity

- Use UUID identifiers.
- Store timestamps in UTC.
- Use indexes for frequently queried columns.
- Enforce uniqueness for external opportunity identifiers where available.
- Use normalized URLs or stable fingerprints for fallback deduplication.
- Use foreign keys and appropriate deletion behavior.
- Preserve score history where it is useful.
- Prevent one user's records from being exposed to another user.
- Validate JSON configuration before saving it.

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11. LEAD SCORING SPECIFICATION

11.1 Score range

Each lead receives a score from 0 to 100.

The score represents prioritization, not the probability of winning a contract.

11.2 Initial formula

Total score:

"0.35 × Relevance + 0.25 × Evidence of Need + 0.15 × Budget Signals + 0.15 × Freshness + 0.10 × Source Confidence"

Each component is normalized to 0–100.

Relevance — 35%

Evaluate:

- Match with offered services.
- Match with demonstrated skills.
- Compatibility with the owner's portfolio.
- Fit with the opportunity's technical requirements.

Evidence of need — 25%

Evaluate:

- Explicitly stated project requirements.
- A published hiring or contracting request.
- A clearly documented business requirement.
- Evidence supporting a possible technical improvement.

A speculative website improvement must not receive the same evidence rating as an explicit request for a developer.

Budget signals — 15%

Evaluate:

- Stated budget.
- Hourly or fixed-price compensation.
- Explicit commercial intent.
- Evidence that the opportunity is paid.

Unknown budget information must be handled transparently and must not automatically disqualify a lead.

Freshness — 15%

Evaluate:

- Publication date.
- Last update.
- Whether the opportunity appears active.
- Whether it has expired or closed.

Do not invent publication dates when unavailable.

Source confidence — 10%

Evaluate:

- Source authenticity.
- Whether the original listing is accessible.
- Whether company details are corroborated.
- Whether important claims have evidence.

11.3 Notification threshold

Default threshold: 65/100.

The owner must be able to change the threshold.

The system should support different thresholds for active job opportunities and proactive business prospects in a future iteration if actual usage demonstrates a need.

11.4 Score explanation

Each score must include:

- Component scores.
- Total score.
- Reason for the match.
- Missing information.
- Evidence links.
- Confidence caveats.
- Recommended next action.

The system must not use an unexplained AI-generated number as the sole basis for prioritization.

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12. AUTOMATED DISCOVERY WORKFLOW

12.1 Scheduled discovery

The scheduled process must execute on a server or supported scheduler independently of the browser.

Recommended initial schedule:

- Active job discovery: every 6–12 hours where source rules and quotas permit.
- Business prospecting: once daily.
- Research and scoring: after a new lead is discovered.
- Email delivery: as soon as a qualifying lead is processed.
- Failed transient tasks: retry with bounded exponential backoff.
- Operational reconciliation: periodically.
- Data cleanup: weekly, subject to retention requirements.

12.2 Processing pipeline

1. Acquire a source-specific execution lock.
2. Load enabled sources and search configurations.
3. Execute permitted searches.
4. Normalize and validate returned records.
5. Reject malformed or unsupported results.

6. Identify duplicates.
7. Create or update lead records.
8. Research new or meaningfully changed leads.
9. Store source evidence.
10. Calculate and persist the lead score.
11. Evaluate notification rules.
12. Create dashboard notification records.
13. Queue email notifications.
14. Record delivery outcomes.
15. Record the final execution status and usage.
16. Release the execution lock.

12.3 Reliability requirements

- A failure in one source must not stop all other sources.
- A task must not create duplicate notifications when retried.
- Re-running a successful task must not duplicate leads.
- Temporary provider errors must use bounded retries.
- Permanent errors must be recorded and surfaced to the administrator.
- Scheduled runs must not overlap in ways that corrupt data.
- The owner must be able to trigger a manual run where authorized.
- Search usage must be capped according to configured limits.

12.4 Free-tier operation

The system must remain usable when:

- A search API is not configured.
- An AI provider is unavailable.
- Email credentials are missing.
- A provider reaches its quota.
- A scheduled execution fails.

In these cases, the application must clearly show the limitation, preserve existing data and support manual imports where appropriate.

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13. INTEGRATIONS

13.1 Supabase

Purpose:

- Authentication.
- PostgreSQL database.
- Row Level Security.
- Server-side functions.
- Scheduled execution support where configured.

Requirements:

- Use migrations.
- Use least-privilege access.
- Protect service-role credentials.
- Document free-tier limitations.
- Provide setup instructions.

13.2 Web search provider

Initial candidate: Brave Search API.

Purpose:

- Find public job announcements.
- Discover companies.
- Identify public business websites.
- Find relevant public pages.

Requirements:

- Use documented endpoints.
- Store source URLs.
- Respect quotas and terms.
- Provide a manual-import fallback.
- Do not assume a free quota is available without completing the provider's requirements.

13.3 Public feeds and job APIs

Integrate permitted sources individually.

Before enabling a source, document:

- Authentication requirements.
- Access permissions.
- Rate limits.
- Attribution obligations.
- Data retention restrictions.
- Whether automated collection is allowed.
- Whether the intended personal-dashboard use is permitted.

Upwork and LinkedIn must not be treated as unrestricted scraping targets. Use permitted access, public search links and user-directed research.

13.4 Resend or equivalent email provider

Purpose: Transactional opportunity notifications.

Requirements:

- Server-side API key.
- Verified sender configuration where required.
- Retry and delivery-status tracking.
- Idempotent sending.
- Email templates with original source links.
- Appropriate unsubscribe and suppression handling if message types or applicable rules require it.

The application should distinguish operational notifications from marketing emails.

13.5 AI model provider

Purpose:

- Extract job requirements.
- Summarize company research.
- Classify opportunities.
- Explain match scores.

- Generate suggested next actions.

Requirements:

- Use a provider abstraction.
- Validate structured output.
- Treat source text as untrusted input.
- Limit token usage and input sizes.
- Avoid sending unnecessary personal data.
- Provide a non-AI fallback.
- Do not assume that a consumer AI subscription includes API credits.

13.6 GitHub

Purpose:

- Version control.
- Private source repository.
- Database migration history.
- Automated tests.
- Optional supported scheduled workflows.

Requirements:

- Protect repository secrets.
- Never commit environment files containing credentials.
- Use pull requests or reviewable commits where practical.
- Do not depend on scheduled actions beyond applicable plan quotas.

13.7 Deployment

Use a suitable free-tier frontend host and Supabase initially.

The chosen deployment must support the required server-side execution, secrets, authentication redirects and scheduled jobs. If the free tier cannot reliably support a requirement, document an alternative rather than silently disabling it.

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14. SECURITY AND PRIVACY REQUIREMENTS

14.1 Authentication security

- Use the authentication provider's supported password handling.
- Do not store plaintext passwords.
- Protect password-reset flows.
- Validate sessions on protected operations.
- Revoke sessions where supported.
- Apply rate limits to sensitive authentication endpoints.
- Enable stronger authentication for privileged accounts where available.

14.2 Authorization security

- Enforce role permissions server-side.
- Apply RLS to user-owned tables.
- Prevent users from modifying their own role.
- Restrict administrative operations to authorized identities.
- Use secure initial-owner provisioning.
- Audit role changes and sensitive configuration changes.

14.3 Secret management

- Keep secrets in server-side environment variables or the provider's secret manager.
- Never expose service-role keys to the browser.
- Never commit secrets to Git.
- Rotate credentials after suspected exposure.
- Do not include secrets in logs or error responses.

14.4 Web data security

- Treat all retrieved content as untrusted.
- Prevent prompt injection from external websites.
- Validate URLs before fetching them.
- Block requests to internal or private network addresses.
- Apply limits to page size, response time and redirects.
- Restrict outbound requests to appropriate protocols and destinations.
- Prevent server-side request forgery.
- Sanitize displayed content to prevent cross-site scripting.
- Do not execute scripts obtained from retrieved pages.

14.5 Privacy and lawful use

- Collect only data needed for legitimate professional prospecting.
- Prefer published business contact information.
- Avoid sensitive personal data.
- Respect applicable privacy and direct-marketing requirements.
- Provide lead deletion and appropriate data-retention controls.
- Maintain source provenance.
- Honor relevant opt-outs and suppression requirements.
- Review applicable Nigerian and international requirements before expanding collection or outreach.

14.6 Operational safeguards

- Apply provider quotas.
- Limit scheduled execution frequency.
- Avoid infinite retry loops.
- Maintain error logs.
- Prevent duplicate notifications.
- Record significant administrative actions.
- Provide a documented recovery procedure.
- Preserve useful functionality when optional services fail.

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15. NON-FUNCTIONAL REQUIREMENTS

15.1 Performance

- Dashboard pages should load efficiently on ordinary mobile connections.
- Use pagination for large lead lists.
- Index commonly filtered database fields.
- Avoid blocking the user interface while research jobs execute.
- Process long-running tasks asynchronously.

Initial performance targets should be measured during testing rather than assumed from development builds.

15.2 Availability and resilience

The application must tolerate temporary outages of search, AI and email providers.

A failure in one external integration should not corrupt lead data or prevent access to existing records.

15.3 Maintainability

- Use typed interfaces.
- Separate business logic from UI components.
- Keep provider integrations modular.
- Use database migrations.
- Document configuration.
- Add automated tests.
- Record architectural decisions where necessary.

15.4 Accessibility

- Use accessible labels and keyboard navigation.
- Provide visible focus states.
- Ensure appropriate text contrast.
- Communicate loading and error states.
- Support screen readers for major controls.

15.5 Scalability

The first release is intended for personal use.

The architecture should support adding sources and additional users without requiring a complete rewrite, but enterprise-scale infrastructure is not an MVP requirement.

15.6 Cost controls

- Prefer free tiers and open-source tools.
- Use rule-based processing when AI is unavailable.
- Limit expensive model calls.
- Cache appropriate research results.
- Track search and AI usage.

- Allow administrators to disable expensive sources.
- Never initiate paid subscriptions or purchases without explicit authorization.

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16. SCHEDULED JOBS AND SYSTEM OPERATIONS

The following background jobs are required or recommended.

Job| Initial schedule| Purpose
Active opportunity discovery| Every 6–12 hours| Find new job posts
Business prospecting| Daily| Discover potential business clients
Research processor| On discovery| Investigate and score leads
Notification dispatcher| As needed| Send qualifying alerts
Failed delivery retry| Periodically| Recover temporary email failures
Source health check| Daily| Detect broken integrations
Usage reconciliation| Daily| Check quotas and execution totals
Data cleanup| Weekly| Apply retention rules

The scheduler must be compatible with the selected hosting and backend services. The exact implementation should be decided after verifying current platform capabilities and limits.

Each scheduled job must have:

- An execution identifier.
- A start time.
- A completion time.
- A status.
- A source or task identifier.
- Counters for processed records.
- Error details where appropriate.
- Retry tracking.
- Protection against overlapping execution.

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17. TESTING REQUIREMENTS

17.1 Unit testing

Test:

- Lead normalization.
- URL normalization.
- Duplicate detection.
- Service matching.
- Score calculations.
- Notification eligibility.
- Search-term generation.
- Configuration validation.
- Status transitions.

17.2 Integration testing

Test:

- Supabase authentication.
- Database migrations.
- RLS policies.
- Source provider responses.
- Scheduled execution.
- Email delivery.
- Notification persistence.
- Failed-task retries.
- AI structured-output validation.

17.3 Security testing

Test:

- Unauthenticated access to private routes.
- Cross-user record access.
- Unauthorized role changes.
- Privilege escalation attempts.
- Secret exposure.
- Malicious external content.
- SSRF attempts.
- XSS attempts.
- Rate-limit enforcement.

- Session revocation.

17.4 End-to-end testing

Verify that:

1. The owner can sign in.
2. The owner can configure services.
3. The application can import a permitted opportunity.
4. The opportunity is saved and deduplicated.
5. Research evidence is stored.
6. The lead receives a score.
7. A qualifying lead creates a dashboard notification.
8. An email alert is sent when the email integration is configured.
9. Failed email delivery does not lose the dashboard alert.
10. The owner can update the lead status.
11. An unauthorized user cannot access privileged pages.
12. A scheduled job can execute when the browser is closed.

17.5 Acceptance evidence

The developer must provide:

- Build output.
- Test results.
- Database migration instructions.
- Deployment instructions.
- A list of configured and unconfigured integrations.
- Evidence that authentication and authorization policies were tested.
- Known limitations.

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18. MVP ACCEPTANCE CRITERIA

The MVP is considered functionally complete only when all critical criteria below have been demonstrated.

Authentication and authorization

- Sign-in and sign-out work.
- Password recovery works.
- Protected routes reject unauthenticated users.
- Super Admin, Admin and User permissions are enforced.
- Unauthorized users cannot grant themselves privileges.
- Database policies prevent cross-user data access.

Discovery

- At least one real, permitted discovery source works.
- Manual URL import works.
- Both active opportunities and business prospecting are supported.
- Search results retain source links.
- Duplicate records are controlled.
- Failed source runs are recorded.

Research and scoring

- Leads can be researched.
- Evidence is linked to its source.
- Scores are calculated and explained.
- Missing information is labeled honestly.
- Scoring works without a paid AI API.

Notifications

- Dashboard notifications persist.
- Email notifications work when configured.
- Both channels can be enabled together.
- Duplicate alerts are prevented.
- Email failure does not erase dashboard alerts.

Administration

- Source configuration is manageable.
- Scheduled runs can be monitored.
- Failed jobs are visible.
- Role changes are protected and audited.
- Secrets remain server-side.

Deployment

- The application works on a phone and laptop.
- Production build succeeds.
- Database migrations are reproducible.
- Setup instructions are complete.
- Free-tier limitations are documented.
- Real integrations are verified rather than simulated.

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19. IMPLEMENTATION ROADMAP

The project will be built incrementally. Estimates assume focused solo AI-assisted development and may change based on integration access and debugging.

Phase 1 — Foundation

Estimated duration: Week 1

Deliverables:

- Project setup.
- Responsive layout.
- Authentication pages.
- Sign-in and sign-out.
- Password recovery.
- Database migrations.
- Initial Super Admin provisioning.
- RLS policies.

Exit criterion: The owner can securely authenticate and access protected pages.

Phase 2 — Administration and lead management

Estimated duration: Week 2

Deliverables:

- Role-based access control.
- Admin dashboard.
- Source management interface.
- Audit logging.
- Lead database and lead detail views.
- Manual opportunity import.

Exit criterion: The owner can manage the application and create and review leads securely.

Phase 3 — Discovery integrations

Estimated duration: Weeks 3–4

Deliverables:

- Public search provider integration where available.
- Business prospecting.
- Active opportunity discovery.
- Normalization and deduplication.
- Source monitoring.
- Scheduled execution.

Exit criterion: At least one real source produces usable leads, and scheduled tasks run independently of the browser.

Phase 4 — Research and scoring

Estimated duration: Week 5

Deliverables:

- Research pipeline.
- Evidence storage.
- AI provider abstraction.
- Rule-based fallback.
- Score calculations.
- Lead ranking and explanations.

Exit criterion: Discovered opportunities can be ranked and the reasoning can be

inspected.

Phase 5 — Notifications

Estimated duration: Week 6

Deliverables:

- Dashboard notification center.
- Email integration.
- Delivery queue.
- Duplicate-send prevention.
- Retry handling.
- Notification preferences.

Exit criterion: Qualifying leads produce persistent dashboard alerts and email alerts when configured.

Phase 6 — Hardening and launch

Estimated duration: Weeks 7–8

Deliverables:

- Security testing.
- Reliability improvements.
- Mobile usability review.
- Deployment.
- Documentation.
- End-to-end verification.
- Initial operational monitoring.

Exit criterion: The owner can use the deployed application for real opportunity discovery with known limitations documented.

These estimates are not guaranteed delivery dates. A minimal version may be usable earlier, while robust multi-source automation may take longer.

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20. SUCCESS METRICS

The platform should measure meaningful outcomes, not just the number of search results.

Discovery metrics

- Opportunities discovered.
- Unique leads created.
- Duplicate rate.
- Leads by source.
- Leads by service category.
- Leads by market.
- Source failure rate.

Quality metrics

- Percentage of leads with valid source links.
- Percentage of leads with usable evidence.
- Percentage of leads matching the owner's services.
- Percentage of leads that remain active when reviewed.
- Percentage of contact details that can be verified.

Operational metrics

- Successful scheduled runs.
- Average processing time.
- Email delivery success rate.
- Notification duplication rate.
- Search quota usage.
- AI API expenditure, if any.

Business outcome metrics

- Leads reviewed.
- Applications submitted.
- Prospects contacted.
- Replies received.
- Interviews or discovery calls booked.
- Projects won.

- Revenue attributed to the platform.

The first month should establish baseline performance. Targets should be based on observed data rather than arbitrary promises about the number of clients the system will generate.

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21. RISKS AND MITIGATIONS

Risk 1: Free sources are insufficient

Mitigation: Start with manual imports, permitted public feeds and limited free search access. Make sources replaceable.

Risk 2: Search results are irrelevant

Mitigation: Improve search terms, add exclusions, score relevance and measure results by source.

Risk 3: AI generates inaccurate information

Mitigation: Require evidence URLs, validate structured output and distinguish facts from hypotheses.

Risk 4: Third-party access restrictions

Mitigation: Review source terms before implementation. Use authorized access methods and original source links.

Risk 5: Free-tier quotas are exceeded

Mitigation: Implement usage counters, configurable limits and a graceful fallback.

Risk 6: Scheduled jobs fail silently

Mitigation: Store run history, expose last-success timestamps and notify the owner about persistent operational failures.

Risk 7: Unauthorized administrative access

Mitigation: Enforce server-side role checks, RLS, MFA where supported and audited role changes.

Risk 8: Duplicate or excessive email notifications

Mitigation: Use idempotency keys, threshold controls and retry limits.

Risk 9: Contact information is missing

Mitigation: Preserve the source URL and use official contact channels where available. Do not fabricate contact details.

Risk 10: The platform becomes too complex

Mitigation: Deliver the smallest reliable vertical slice first, then add integrations based on evidence of value.

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22. INITIAL DEPLOYMENT AND OPERATING BUDGET

The initial target is $0.

Component| Initial approach| Target cost
Frontend hosting| Suitable free tier| $0
Database and authentication| Supabase Free| $0
Source control| GitHub Free| $0
Email notifications| Suitable free transactional tier| $0
Search| Manual import and permitted free sources| $0
AI research| Rule-based fallback until free access is available| $0
Custom domain| Use a provider subdomain| $0

This budget is conditional on current provider terms, quotas, eligibility and account requirements. It does not guarantee unrestricted automated searches or unlimited background execution.

The platform must never require an unexpected purchase to continue basic operation. If a provider requires payment information or paid access, the integration must remain optional until the owner explicitly enables it.

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23. RELEASE STRATEGY

Internal alpha

The owner uses the app to import a small set of real opportunities and verifies authentication, research, scoring and notifications.

Personal beta

Enable one or more permitted discovery sources. Observe lead quality, search failures, duplicate rates and notification usefulness.

MVP release

Enable scheduled discovery, dual notifications, administrative controls and operational monitoring. Begin tracking whether discovered leads produce real conversations.

Future expansion

Only after the MVP demonstrates value, consider:

- More discovery integrations.
- Additional permitted job feeds.
- More sophisticated company research.
- Source-specific score adjustments.
- Better prioritization based on actual outcomes.
- Optional outreach-draft generation.
- Advanced analytics.
- Additional user accounts.
- More advanced notification rules.

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24. FINAL PRODUCT DECISIONS

The following decisions define the initial product and should be treated as the

baseline for implementation.

1. The product name is Remy Prospect AI.
2. It will be a separate application, not a module inside the existing Remy AI application.
3. It will be a responsive web application usable on phones and laptops.
4. Nigerian and international clients will receive equal discovery priority.
5. Active freelance opportunities and proactive business prospecting will receive equal priority.
6. Initial service categories will include AI-assisted websites, AI-assisted mobile applications, API integrations, AI solutions and related development work.
7. The assistant will search automatically where permitted.
8. It will research, deduplicate, rank and score opportunities.
9. It will notify the owner through both email and the in-app dashboard.
10. The default lead-notification threshold will be 65/100.
11. The owner will have a Super Admin account established through secure provisioning.
12. The application will support Super Admin, Admin and User roles.
13. Authentication will include sign-in, sign-out, password recovery and protected routes.
14. The initial architecture will use React, TypeScript, Supabase and modular integrations.
15. The project will prioritize free tiers and useful fallback behavior.
16. Unauthorized scraping, private contact extraction and platform-access circumvention are prohibited.
17. Automated research and notifications will not require approval for every individual lead.
18. External outreach will remain under the owner's control in the MVP.
19. The system will preserve source evidence and explain its scores.
20. The implementation will be accepted only after its core workflows and security controls have been tested.

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25. DEFINITION OF PRODUCT SUCCESS

Remy Prospect AI will have achieved its initial purpose when the owner can open the application and find a continuously maintained, evidence-backed list of relevant development opportunities, including opportunities discovered while the owner was offline.

Each useful lead should answer:

- What is the opportunity?
- Who is the prospective client?
- What service might they need?
- Where is the original evidence?
- How well does it match the owner's skills?
- What is known and what remains uncertain?
- How can the owner pursue it legitimately?
- What action should happen next?

The product's ultimate success will not be measured by how many AI features it contains or how many websites it claims to search. It will be measured by whether it consistently helps the owner discover genuine opportunities, make better decisions, build professional relationships and win paid development work.

End of Product Requirements Document.